

Lab: Blockchain

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Lab: 7

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Task 1

Sepolia Testnet

Search by Address / Txn Hash / Block / Token

Etherscan

HomeBlockchainTokensNFTsMore

Address

0x643743E3923f15fb07BcC83CC34D3c41A30357E7

Overview

ETH BALANCE
0.284 ETH

More Info

TRANSACTIONS SENT
Latest: N/A First: N/A

FUNDED BY
0x6Cc9397c...77Ba5F455 | 3 mins ago

Multichain Info

N/A

Transactions

Token Transfers (ERC-20)

Latest 1 from a total of 1 transactions

Download Page Data

Transaction Hash	Method	Block	Age	From	To	Amount	Txn Fee
0xc9ab7d47c0...	Transfer	10939407	2 mins ago	0x6Cc9397c...77Ba5F455	0x643743E3...1A30357E7	0.284 ETH	0.00038951

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Powered by Ethereum

Add Sepolia Test Network

Etherscan © 2026 (Sepolia)

Terms & Privacy | Network Status | Donations: 0x71c765...d8976f

Task 2

```
// SPDX-License-Identifier: MIT
pragma solidity >=0.8.2 <0.9.0;

// Import Chainlink's AggregatorV3Interface to fetch real-world data
import {AggregatorV3Interface} from
"@chainlink/contracts/src/v0.8/shared/interfaces/AggregatorV3Interface.sol"
;

contract DonationContract {
    // State variables
    address payable public owner;

    // Map address to stored value (keeping track of depositors)
    mapping(address => uint256) public identitiesToAmountFunded;

    // Chainlink Data Feed Interfaces
    AggregatorV3Interface internal ethUsdFeed;
    AggregatorV3Interface internal eurUsdFeed;
```

```

// The constructor runs once when the contract is deployed
constructor() {
    // The first person to deploy the contract is the owner
    owner = payable(msg.sender);

    // Initialize the two different interfaces for price feeds (Sepolia
Testnet Addresses)
    // Note: Check chainlink docs for the most up-to-date Sepolia feed
addresses
    ethUsdFeed =
AggregatorV3Interface(0x694AA1769357215DE4FAC081bf1f309aDC325306);
    eurUsdFeed =
AggregatorV3Interface(0x1A81AFb8146AEFFCdC5E50ce46482eDcc2137ce2);
}

// Function to calculate the current ETH to EUR rate
function getEthEurRate() public view returns (uint256) {
    (, int256 ethUsdPrice, , , ) = ethUsdFeed.latestRoundData();
    (, int256 eurUsdPrice, , , ) = eurUsdFeed.latestRoundData();

    // Calculate ETH/EUR price. Both USD feeds have 8 decimals, so they
cancel out.
    // We multiply by 1e18 to maintain precision for Wei conversions.
    uint256 ethEurPrice = uint256((ethUsdPrice * 1e18) / eurUsdPrice);
    return ethEurPrice;
}

// Payable function to accept donations
function donate() public payable {
    // Get the current ETH/EUR rate
    uint256 ethEurRate = getEthEurRate();

    // Calculate the value of the donation in Euros (with 18 decimal
precision)
    uint256 donationValueInEur = (msg.value * ethEurRate) / 1e18;

    // Require minimum 1 Euro donation (represented as 1e18 because of
decimals)
    require(donationValueInEur >= 1e18, "Minimum donation is 1 Euro");

    // Add entries to mapping to keep track of the depositors
identitiesToAmountFunded[msg.sender] += msg.value;
}

// Function to withdraw funds
function withdraw() public {
    // Allow withdraw only for contract creator
    require(msg.sender == owner, "Only the owner can withdraw funds");

    // Getting balance of the deployed contract
    uint256 contractBalance = address(this).balance;

    // Updated withdrawal method using call

```

```
}  
  
    (bool success, ) = owner.call{value: contractBalance}("");  
    require(success, "Transfer failed.");  
}
```

Sepolia Testnet

Search by Address / Txn Hash / Block / Token



Address 0x643743E3923f15fb07BcC83CC34D3c41A30357E7

Overview

ETH BALANCE

0.000218176635159 ETH

More Info

TRANSACTIONS SENT

Latest: 4 mins ago First: 20 mins ago

FUNDED BY

0x6Cc9397c...77Ba5F455 | 1 hr ago

Multichain Info

N/A

Transactions

Token Transfers (ERC-20)

Latest 8 from a total of 8 transactions

Download Page Data

Transaction Hash	Method	Block	Age	From	To	Amount	Txn Fee
0x06aa647c83...	Donate	10939647	3 mins ago	0x643743E3...1A30357E7	0x727bDAec...a598a56e9	0.05 ETH	0.00052611
0x220972d554...	0x60806040	10939645	4 mins ago	0x643743E3...1A30357E7	Contract Creation	0 ETH	0.00798867
0xd8a0f6993a8...	0x60806040	10939643	4 mins ago	0x643743E3...1A30357E7	Contract Creation	0.049999999 ETH	0.00151395
0xdc9529046e...	0x60806040	10939633	7 mins ago	0x643743E3...1A30357E7	Contract Creation	0.05 ETH	0.00140436
0x6b442b1172...	Donate	10939632	7 mins ago	0x643743E3...1A30357E7	0x5d8a4449...875402A93	0.05 ETH	0.00051859
0x5a8fcb1392c...	Transfer	10939632	7 mins ago	0x643743E3...1A30357E7	SELF 0x643743E3...1A30357E7	0 ETH	0.00022626
0x9b5f9391a75...	0x60806040	10939576	19 mins ago	0x643743E3...1A30357E7	Contract Creation	0 ETH	0.00878424
0xc9ab7d47cd...	Transfer	10939407	57 mins ago	0x6Cc9397c...77Ba5F455	0x643743E3...1A30357E7	0.284 ETH	0.00038951